

# Run Claude Code on Cheap Third-Party Models (and Survive the Token Bill)

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*Tested on Claude Code 2.1.x (Linux). No hacks, no spoofing — just documented config.*

## Why

Claude Code is a great agent loop, but running every turn — plus background tasks, subagents, and retries — on flagship Claude models burns tokens fast. The trick: **keep Claude Code's interface, swap the engine**. Point it at cheaper OpenAI-compatible providers (Nvidia NIM, above.dev, Ollama, OpenCode Zen) through a local translator proxy. Same `/model` picker, same `/tasks` panel, fraction of the cost. Use flagship models for hard reasoning, cheap ones for everything else.

## How it works

Claude Code speaks the **Anthropic Messages API**. Most cheap providers speak **OpenAI's API**. [LiteLLM](#) (open source) sits between them on your own machine and translates both ways:

```
Claude Code  --->  LiteLLM :4000  --->  above.dev / Nvidia / Ollama / Zen
(Anthropic)      (translates)          (OpenAI)
```

Nothing runs models locally (unless you choose Ollama) — the proxy is just a translator, ~250 MB RAM.

## Step-by-step

### 1. Install the proxy (pin a safe version!)

```
uv tool install 'litellm[proxy]'
litellm --version
```

⚠ Versions **1.82.7 / 1.82.8** were **compromised** (credential-stealing malware, BerriAI/litellm#24518). Verify your version isn't one of those.

## 2. Write the gateway config ( ~/.config/claude-gateway/litellm\_config.yaml )

```
general_settings:
  master_key: sk-local-only          # any string; only used on localhost

model_list:
  - model_name: claude-gw-my-cheap-one  # MUST contain "claude" or "anthropic"
    litellm_params:                    # ...or Claude hides it from /model!
      model: hosted_vllm/<provider-model-id>
      api_base: https://<provider>/v1
      api_key: os.environ/MY_PROVIDER_KEY
```

Two gotchas I hit, so you don't have to: - **hosted\_vllm/** prefix, not **openai/**. Newer LiteLLM routes **openai/\*** to the **/v1/responses** endpoint, which most third-party providers don't implement (HTTP 404 with an HTML page). **hosted\_vllm/\*** forces classic **/chat/completions**. - **Model IDs must contain claude or anthropic** (case-insensitive) or Claude's gateway discovery silently filters them out of the **/model** picker. I use a **claude-gw-** prefix. Inference via **--model <id>** works regardless — only the picker is filtered.

## 3. Run it persistently (systemd user service survives reboots)

~/.config/systemd/user/claude-gateway.service:

```
[Unit]
Description=LiteLLM gateway for Claude Code
After=network-online.target

[Service]
Type=simple
ExecStart=%h/.local/bin/litellm --config %h/.config/claude-gateway/
litellm_config.yaml --host 127.0.0.1 --port 4000
Restart=on-failure
```

```
systemctl --user daemon-reload
systemctl --user enable --now claude-gateway.service
curl -s http://127.0.0.1:4000/v1/models -H "Authorization: Bearer sk-local-only"
```

## 4. Point Claude Code at it ( ~/.claude/settings.json )

```
{
  "model": "claude-gw-my-cheap-one",
  "env": {
    "ANTHROPIC_BASE_URL": "http://127.0.0.1:4000",
    "ANTHROPIC_AUTH_TOKEN": "sk-local-only",
    "ANTHROPIC_API_KEY": "",
    "ANTHROPIC_MODEL": "claude-gw-my-cheap-one",
```

```

"ANTHROPIC_DEFAULT_OPUS_MODEL": "claude-gw-my-cheap-one",
"ANTHROPIC_DEFAULT_SONNET_MODEL": "claude-gw-my-cheap-one",
"ANTHROPIC_DEFAULT_HAIKU_MODEL": "claude-gw-my-cheap-one",
"CLAUDE_CODE_ENABLE_GATEWAY_MODEL_DISCOVERY": "1"
}
}

```

Notes: - `ANTHROPIC_API_KEY: ""` (empty) is deliberate — a set key takes a different auth path and can silently bypass your gateway back to Anthropic billing. - Keep `ANTHROPIC_DEFAULT_HAIKU_MODEL` on a cheap model: background chores (titles, summaries) run on it constantly. - Discovery populates `/model` at session startup; restart Claude after changes.

## 5. Verify

```

claude doctor # should mention custom ANTHROPIC_BASE_URL, no install issues
claude -p "Reply with exactly: gateway-ok" --output-format text

```

The `[claude-code:unrecognized_model]` notice is benign — it just means your custom ID isn't in Claude's catalog (it assumes a 200k context; append `[1m]` or set `CLAUDE_CODE_MAX_CONTEXT_TOKENS` if your model allows more).

## Where to get cheap models

| Source                                                                                                                     | Notes from real testing                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Nvidia NIM</b><br>( <code>integrate.api.nvidia.com/v1</code> )                                                          | Big catalog, but verify each model: some are retired (410), some need per-account activation (404 "not found for account"), some are slow. Probe before trusting.                                                                |
| <b>above.dev</b> ( <code>api.above.dev/v1</code> )                                                                         | Keys can be scoped to a single model — check which IDs your key allows. Note: its <code>/v1/messages</code> returns OpenAI-shaped JSON, so it still needs LiteLLM in front.                                                      |
| <b>Ollama local</b><br>( <code>http://localhost:11434</code> , no proxy needed since v0.14 — it speaks Anthropic natively) | <code>ANTHROPIC_BASE_URL=http://localhost:11434</code> , <code>ANTHROPIC_AUTH_TOKEN=ollama</code> , <code>ANTHROPIC_API_KEY=""</code> . Zero token cost, needs RAM/disk.                                                         |
| <b>OpenCode Zen</b> (paid key)                                                                                             | Works great through the gateway. ⚠️ The <b>free</b> Zen tier is server-gated to requests from inside OpenCode itself ( <code>FreeTierError</code> otherwise) — no header/key replay fixes that. Don't go down the spoofing path. |

## The cost playbook (what actually saves money)

1. **Default cheap, escalate on demand.** Daily driver = cheap model; `claude --model opus` (or picker) for the 10% of tasks that need flagship reasoning.
2. **Push background work to the cheapest model** via `ANTHROPIC_DEFAULT_HAIKU_MODEL` — subagents, titles, and summaries are pure token overhead.
3. **Revert in seconds.** Remove the `env` block (keep a backup of settings.json) and you're back on your Anthropic subscription. I flip back and forth regularly.

## Reverting

```
systemctl --user stop claude-gateway.service # optional
# remove the "env" block from ~/.claude/settings.json, restore "model"
claude doctor
```

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*If this saved you money, share it with one developer who's afraid to look at their token bill. 🌊*